

Micromouse

Final Project — Calculations & Justification

IEEE ZSB · RAS Chapter — PCB Design Track Final Project

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PCB Copper Trace Width Calculations

Calculated using the **IPC-2221** standard for external (top/bottom layer) traces, 1oz/ft² copper, 10°C allowable temperature rise.

$$\text{Area (mils}^2\text{)} = (I / (k \times \Delta T^{0.44}))^{(1/0.725)} \quad | \quad \text{Width (mils)} = \text{Area} / 1.37$$

I = current (A), $\Delta T = 10^\circ\text{C}$, $k = 0.048$.

1.1 — Top Floor (Sensors)

Net	Worst-Case I	Calc. Min.	Implemented	Margin
3.3V / GND (power)	0.139 A	0.020 mm	0.3533 mm	17.7×
Signal traces (SDA, SCL, XSHUT)	< 10 mA	< 0.01 mm	0.25 mm	Ample

1.2 — Bottom Floor (Power, MCU, Motor Driver)

Net	Worst-Case I	Calc. Min.	Implemented	Margin
Main power rail (Fuse → Switch → Protection → VCC)	0.789 A	0.218 mm	0.4 mm	1.83×
DRV8833 VM	0.600 A	0.149 mm	0.4 mm	2.68×
Motor outputs → Motor Left / Right (each)	0.300 A	0.057 mm	0.25 mm	4.39×
Regulator path (7.4V_Safe → AMS1117 → 5V → ESP32)	0.189 A	0.030 mm	0.4 mm	13.3×
Signal traces (motor control, encoders, ADC divider)	< 10 mA	< 0.01 mm	0.254 mm	Ample

Calculated using $P = V \times I$ at each component's actual operating voltage.

Component	V	I (typ.)	P	Source
ESP32-C3 (active, radios off)	3.3V	45 mA	0.149W	Espressif datasheet
3× VL53L0X ToF (30mA ea.)	3.3V	90 mA	0.297W	ST datasheet
MPU6050 IMU	3.3V	3.6 mA	0.012W	InvenSense datasheet
OLED 0.96in (SSD1306)	3.3V	20 mA	0.066W	Manufacturer datasheet
2× Encoders (10mA ea., estimated)	3.3V	20 mA	0.066W	Estimated
DRV8833 quiescent	3.3V	5 mA	0.017W	Typical
AMS1117-5.0 quiescent	7.4V	5 mA	0.037W	Typical
Battery divider bleed (100k+33k)	7.4V	0.056 mA	0.0004W	Calculated
Q1 (IRF9540) loss — idle	—	@ 0.189A	0.007W	Rds(on)=0.20Ω
Q1 (IRF9540) loss — typical	—	@ 0.439A	0.039W	Rds(on)=0.20Ω
Q1 (IRF9540) loss — worst-case	—	@ 0.789A	0.125W	Rds(on)=0.20Ω
2× N20 motors — typical (estimated)	7.4V	250 mA	1.850W	~40% of stall
2× N20 motors — worst-case stall	7.4V	600 mA	4.440W	Manufacturer datasheet

System Totals

Every regulation stage (AMS1117, Super Mini onboard regulator) is linear, not switching — current is approximately conserved through each stage rather than reduced, so every downstream component's current is reflected directly in what the battery must supply.

Scenario	Total Current	Total Power	Q1 Loss Share
Logic + sensors only (motors idle)	189 mA	1.399 W	0.5%
Typical driving (motors normal)	439 mA	3.249 W	1.2%
Worst-case (both motors stalled)	789 mA	5.839 W	2.1%

Chosen microcontroller: ESP32-C3 Super Mini

Required: 4 motor-driver control pins, 4 quadrature encoder channels, a 2-pin I2C bus shared by 5 devices, 2 GPIO-controlled ToF address lines, and 1 ADC pin for battery sensing — **13 GPIOs total**.

3.1 — ESP32-C3 Super Mini vs. a Standard 30/38-pin ESP32 DevKit

Spec	ESP32-C3 Super Mini	Standard 30/38-pin DevKit
Board footprint	22.5 × 18 mm = 405 mm ²	~55 × 28 mm = 1,540 mm ² (~3.8× larger)
CPU	Single-core RISC-V	Dual-core Xtensa LX6 (higher idle draw)
Onboard extras	Small onboard 3.3V LDO only	USB-UART bridge chip + large USB connector + full regulator
Usable GPIOs	13 (exact match)	30-38 (17-25 unused in this design)
Mounting	Pluggable, small 2.54mm header	Larger socket footprint or permanent solder-down

- 405mm² footprint is ~3.8× smaller than a standard DevKit's ~1,540mm² — a direct, quantifiable contribution to the 11×11cm limit and smallest-robot bonus
- A DevKit's onboard USB-UART bridge and full-size USB connector exist only for plug-and-program convenience — functionality this design moves off-board by programming the Super Mini externally before mounting, making that circuitry pure dead weight here
- 13 GPIOs is an exact match to this design's requirement; a 30/38-pin DevKit would leave 17-25 pins permanently unused
- Both chip families include WiFi/BLE by default; since this design uses neither, the smaller/cheaper C3 package is strictly better with no functional trade-off

3.2 — Handling the Tight Pin Budget Safely

- GPIO9 (BOOT button pin) carries only a motor-driver output signal, never an encoder input — a driver output is always actively set by firmware and cannot glitch the boot sequence the way an unpredictable encoder state could
- GPIO8 (onboard status LED) is assigned to a non-critical encoder channel — the LED flickering with wheel rotation is cosmetic only

Final Battery Specification

Spec	Value
Type / configuration	2× 3.7V LiPo cells (ECOCELL) in series = 7.4V nominal
Capacity	1000 mAh
Dimensions (per cell)	35 × 25 × 10 mm
Weight (per cell / pair)	22g / 44g total
Max charge current	500 mA (0.5C) — per cell, charged individually
Max discharge current	1000 mA (1C) max; 500 mA (0.5C) recommended continuous
Protection	Built-in per-cell: overcharge, over-discharge, short-circuit

Step 1 — Runtime at Typical Operating Current

$$\text{Runtime (h)} = \text{Capacity (Ah)} / \text{Current (A)}$$

Runtime = 1.0 Ah / 0.439 A ≈ **2.28 hours (~137 minutes)**.

Step 2 — Runtime at Worst-Case Current

Runtime = 1.0 Ah / 0.789 A ≈ **1.27 hours (~76 minutes)** — both motors permanently stalled; pessimistic-case only.

Step 3 — Discharge Rate Check

This battery specifies fixed discharge limits: 500mA recommended continuous, 1000mA absolute maximum. Typical current (439mA) stays under the recommended limit; worst-case current (789mA) exceeds it by 58% but remains under the 1000mA absolute maximum — acceptable for this short-duration competition project.

Step 4 — Practical Safety Margin for Competition Use

$$\text{Required capacity} = \text{Current} \times \text{Runtime needed} \times 1.25$$

For a 15-minute session: $0.439\text{A} \times 0.25\text{h} = 0.110\text{ Ah}$, $\times 1.25 \approx$ **137 mAh minimum**. The chosen 1000mAh pack provides over 7× this minimum.

Step 5 — Charging Procedure

Two independently-protected single cells, not one factory 2S pack — charge each cell **individually**: disconnect the series link, charge each 3.7V cell separately through its own JST-PH connector at or under 500mA, then reconnect in series for use.